When a potential difference V is applied across the wire, a net electric field is created in the wire which constitutes the current in the wire. For simplicity, we assume that the electric field is uniform in the entire length of the wire, then the potential difference (voltage V) can be written as

$V = El$

As we know, the magnitude of current density

$$J = \sigma E = \sigma \frac{V}{l} \qquad (2.14)$$

But $J = \frac{I}{A}$, so we write the equation (2.14) as

$$\frac{I}{A} = \sigma \frac{V}{l}.$$

By rearranging the above equation, we get

$$V = I\left(\frac{l}{\sigma A}\right) \qquad (2.15)$$

The quantity $\frac{l}{\sigma A}$ is called resistance of the conductor and it is denoted as R. Note that the resistance is directly proportional to the length of the conductor and inversely proportional to area of cross section.

Therefore, the macroscopic form of Ohm's law can be stated as "**the potential difference across a given conductor is directly proportional to the current passing throught it when the temperature remains constant**".

$$V = IR \qquad (2.16)$$

From the above equation, **the resistance is the ratio of potential difference across the given conductor to the current passing through the conductor.**

$$R = \frac{V}{I} \qquad (2.17)$$

The SI unit of resistance is ohm (Ω). From the equation (2.16), we infer that the graph between current versus voltage is straight line with a slope equal to the inverse of resistance R of the conductor. It is shown in the Figure 2.8 (a).

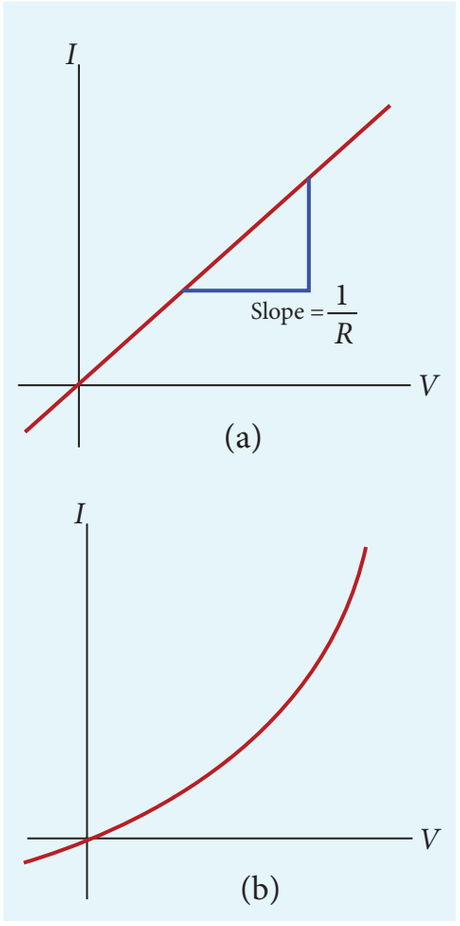


Figure 2.8 Current against voltage for (a) a conductor which obeys Ohm's law and (b) for a non-ohmic device (Diode given in XII physics, unit 10 is an example of a non-ohmic device)

Materials for which the current versus voltage graph is a straight line through the origin, are said to obey Ohm's law and their behaviour is said to be ohmic as shown in Figure 2.8(a). Materials or devices that do not follow Ohm's law are said to be non-ohmic. These materials have more complex relationships between voltage and current. A plot of I versus V for a non-ohmic material is non-linear and they do not have a constant resistance (Figure 2.8(b)).

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